

Integer Programming for Large Independent Sets in Sparse Random Networks and ca-GrQc Real Network

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Abstract

We study the maximum independent set problem on sparse Erdős–Rényi graphs $G(n, d/n)$ and on the SNAP ca-GrQc collaboration network. We are interested not only in the size of the independent sets returned by a warm-started integer program, but also in whether the solver certifies optimality. On $G(n, 20/n)$, a multi-seed experiment over sizes $n = 100$ through 5000 shows random-order greedy closely following the finite- d prediction $\log(1 + d)n/d$, while minimum-degree greedy gives substantially larger warm starts for CBC. CBC certifies all $n = 100$ random instances, but every larger random instance in this experiment ends without a matching upper bound; for $n \geq 500$, the final incumbent is just the minimum-degree warm start. Thus, on the larger random instances, the limiting issue is certification rather than the construction of a large feasible set. By contrast, on the real ca-GrQc network, connected-component decomposition makes the same IP formulation effective as an exact method, allowing CBC to *certify* $\alpha(G) = 2459$, while minimum-degree greedy returns 2458. The contrast suggests that the same IP formulation is limited less by vertex count alone than by graph structure, with the sparse random graphs leaving CBC with weak global certificates and ca-GrQc splitting into many components and containing collaboration cliques that make exact certification tractable.

1 Introduction

The maximum independent set problem asks for the largest set of vertices in a graph with no edges among the selected vertices. In this project, we evaluate an integer-programming formulation for this problem on two classes of sparse graphs, namely Erdős–Rényi random graphs $G(n, d/n)$ [1], with particular attention to the case $d = 20$, and the SNAP ca-GrQc collaboration network, a real graph with clustering, heterogeneous degrees, and many connected components.

The asymptotic scale of maximum independent sets in $G(n, d/n)$ is well known [2], with the independence number in the double asymptotic regime $n \rightarrow \infty$ followed by $d \rightarrow \infty$ being on the order of

$$\frac{2 \log d}{d} n,$$

whereas simple greedy procedures achieve only about

$$\frac{\log d}{d} n.$$

The question is therefore whether integer programming can meaningfully *exceed* the random-order greedy scale, and whether it can provide optimality *certificates* at the graph sizes considered here.

Our results separate the quality of feasible solutions from the ability to prove optimality. On the tested random graphs, the warm-started IP pipeline returns feasible independent sets above the

random-order greedy scale, but the solver usually cannot prove that these incumbents are optimal. In the multi-seed random-graph experiment, CBC certifies optimality for all $n = 100$ instances, but for $n \geq 200$ it terminates with a nonzero certificate gap. On ca-GrQc, after decomposing the graph into connected components, the IP formulation instead certifies the exact value $\alpha(G) = 2459$.

2 IP Formulation

For a graph $G = (V, E)$, an independent set is a subset $S \subseteq V$ such that no two vertices in S are adjacent. The independence number is

$$\alpha(G) = \max\{|S| : S \subseteq V, S \text{ independent}\}.$$

The standard binary integer-programming formulation uses a variable $x_v \in \{0, 1\}$ for each vertex and solves

$$\begin{aligned} \max \quad & \sum_{v \in V} x_v \\ \text{s.t.} \quad & x_u + x_v \leq 1, \quad (u, v) \in E, \\ & x_v \in \{0, 1\}, \quad v \in V. \end{aligned}$$

The edge constraints ensure that adjacent vertices cannot both be selected, so the feasible binary vectors are precisely the incidence vectors of independent sets.

In the computations we solve the equivalent minimum vertex-cover formulation. With the change of variables $y_v = 1 - x_v$, the model becomes

$$\begin{aligned} \min \quad & \sum_{v \in V} y_v \\ \text{s.t.} \quad & y_u + y_v \geq 1, \quad (u, v) \in E, \\ & y_v \in \{0, 1\}, \quad v \in V. \end{aligned}$$

If $\tau(G)$ denotes the minimum vertex-cover value, then

$$\alpha(G) = |V| - \tau(G).$$

In our computations, CBC was more stable on this cover formulation once warm starts were enabled.

For instances that are not solved to optimality, we report both incumbent quality and the remaining certificate gap. Let L denote the best feasible independent set size found by the solver. In the vertex-cover formulation, CBC also maintains a lower bound B_{VC} on the optimal cover size, which gives the independent-set upper bound

$$U = |V| - \lceil B_{VC} \rceil.$$

Thus

$$\alpha(G) \in [L, U].$$

We also record the warm-start size W and the improvement

$$\Delta = L - W.$$

The value of Δ is a useful diagnostic because it indicates whether CBC actually improved on the greedy solution used to initialize the solve.

3 Theory Benchmarks

The leading asymptotic expression $2\log(d)n/d$ gives the correct scale when d is large, but it is not a sharp numerical prediction at $d = 20$, so we compare the experiments against the random-order greedy prediction and a first-moment upper benchmark as two finite- d reference quantities.

3.1 Greedy Scale

Consider random-order greedy, which processes vertices in a uniformly random order and accepts a vertex whenever it has no previously accepted neighbor. Let $s(t)$ denote the accepted fraction after a fraction t of vertices has been processed. For $G(n, d/n)$, a standard differential-equation calculation in the style of Wormald's method for random graph processes [3] gives

$$\frac{ds}{dt} = \exp(-ds(t)), \quad s(0) = 0.$$

A newly exposed vertex is accepted when it has no edge into the already accepted set, whose size is approximately $s(t)n$. Solving the differential equation yields

$$s(t) = \frac{\log(1 + dt)}{d}, \quad s(1) = \frac{\log(1 + d)}{d}.$$

For $d = 20$,

$$\frac{\log(21)}{20} = 0.1522,$$

which is close to the threshold

$$\frac{\log(20)}{20} = 0.1498.$$

Accordingly, $\log(1 + d)n/d$ and $\log(d)n/d$ are both useful benchmarks for the greedy scale in our experiments.

3.2 First-Moment Upper Benchmark

A finite- d first-moment calculation provides a sharper upper reference than the leading large- d expression. Let N_k be the number of independent sets of size $k = an$. Then

$$\mathbb{E}[N_k] = \binom{n}{k} (1 - d/n)^{\binom{k}{2}}.$$

By Stirling's formula,

$$\frac{1}{n} \log \mathbb{E}[N_{an}] = H(a) - \frac{da^2}{2} + o(1),$$

where

$$H(a) = -a \log a - (1 - a) \log(1 - a).$$

If

$$H(a) - \frac{da^2}{2} < 0,$$

then Markov's inequality rules out independent sets of size an with high probability. Thus the largest positive root of

$$H(a) - \frac{da^2}{2} = 0$$

serves as a finite- d upper benchmark. At $d = 20$, this root is

$$a_{\text{FM}} = 0.232996.$$

This is substantially below

$$\frac{2 \log(20)}{20} = 0.2996,$$

so the first-moment benchmark is the more informative reference for the moderate-degree experiments reported below.

4 Implementation

We implemented and compared three procedures.

- **Random-order greedy** processes vertices in random order and accepts each vertex whenever possible. We used 20 random permutations in the low-time-limit comparison and 100 random permutations per instance in the multi-seed random-graph experiment.
- **Minimum-degree greedy** selects a vertex of current minimum degree at each step, after which the selected vertex together with its neighbors is removed from the graph.
- **IP/CBC** solves the binary vertex-cover formulation in PuLP 3.3.0 using CBC 2.10.3, initialized with the best greedy solution available [4].

For each Erdős–Rényi instance, we generated a graph from $G(n, 20/n)$, ran the greedy baselines, used the best greedy solution as a warm start, and then solved the vertex-cover IP subject to the stated time limit. The main random graph experiment uses 20 seeds each for $n = 100, 200, 500, 1000$, 10 seeds each for $n = 1500, 2000, 3000$, and 5 seeds for $n = 5000$, with a 300-second CBC limit for $n \leq 1000$ and a 600-second limit for the larger scaling sizes. We also use a single-seed low-time-limit comparison, with limits of 30 seconds at $n = 100$, 20 seconds for $200 \leq n \leq 2000$, and 60 seconds at $n = 3000$. For each run we recorded the warm-start size W , final incumbent L , improvement $\Delta = L - W$, converted upper bound U , and whether CBC certified optimality.

For the ca-GrQc experiment, we first converted the SNAP edge list into a simple undirected graph by symmetrizing duplicate directed rows, removing self-loops, and retaining vertices that became isolated after loop removal, then split the resulting graph into connected components. Since independence number is additive over connected components,

$$\alpha(G) = \sum_i \alpha(G_i),$$

we solved the IP separately on each component and summed the resulting optima.

We also included clique inequalities in the random-graph IP computations. In the independent-set variables, every clique C yields

$$\sum_{v \in C} x_v \leq 1.$$

In the vertex-cover variables $y_v = 1 - x_v$, this becomes

$$\sum_{v \in C} y_v \geq |C| - 1.$$

We used triangle and 4-clique cuts for the random graphs. These cuts were not used for ca-GrQc; after component decomposition, the edge formulation was already sufficient for the real-network instance.

All returned independent sets were verified directly against the edge list. As a separate sanity check, small validation instances were solved both by CBC and by exhaustive search. The two methods agreed on

$$(n, d, \alpha) = (12, 3, 6), \quad (14, 4, 5), \quad (16, 5, 7).$$

All experiments were run on an Apple M3 Pro with 18 GB RAM, Python 3.9.6, PuLP 3.3.0, and CBC 2.10.3. The notebook and supporting code are available in the project repository [7].

5 Random Graph Experiments

For the Erdős–Rényi part of the project, we use the completed multi-seed experiment as the main source of evidence rather than relying on a single random seed. It uses 20 seeds each for $n = 100, 200, 500, 1000$, 10 seeds each for $n = 1500, 2000, 3000$, and 5 seeds for $n = 5000$. Within each instance, random-order greedy is averaged over 100 permutations, the best of random-order greedy and minimum-degree greedy is used as the IP warm start, and CBC is then asked both to improve the incumbent and to prove an upper bound. We used a 300-second limit for $n \leq 1000$ and a 600-second limit for the larger scaling sizes.

Table 1: Multi-seed summary for $G(n, 20/n)$. Ratios are means over seeds. Here L is the incumbent independent-set size and U is CBC’s converted upper bound on $\alpha(G)$. The certified and uncertified columns count how many instances did or did not receive an optimality certificate.

n	graphs	certified	uncertified	rand./ n	min-deg/ n	L/n	U/n	time/graph
100	20	20	0	0.1442	0.1830	0.1990	0.1990	14.3s
200	20	0	20	0.1485	0.1888	0.1963	0.2513	300.6s
500	20	0	20	0.1508	0.1924	0.1924	0.3219	290.7s
1000	20	0	20	0.1515	0.1931	0.1931	0.3552	161.2s
1500	10	0	10	0.1515	0.1955	0.1955	0.3678	599.6s
2000	10	0	10	0.1519	0.1965	0.1965	0.3760	596.9s
3000	10	0	10	0.1521	0.1953	0.1953	0.4000	403.5s
5000	5	0	5	0.1522	0.1958	0.1958	0.4293	596.3s

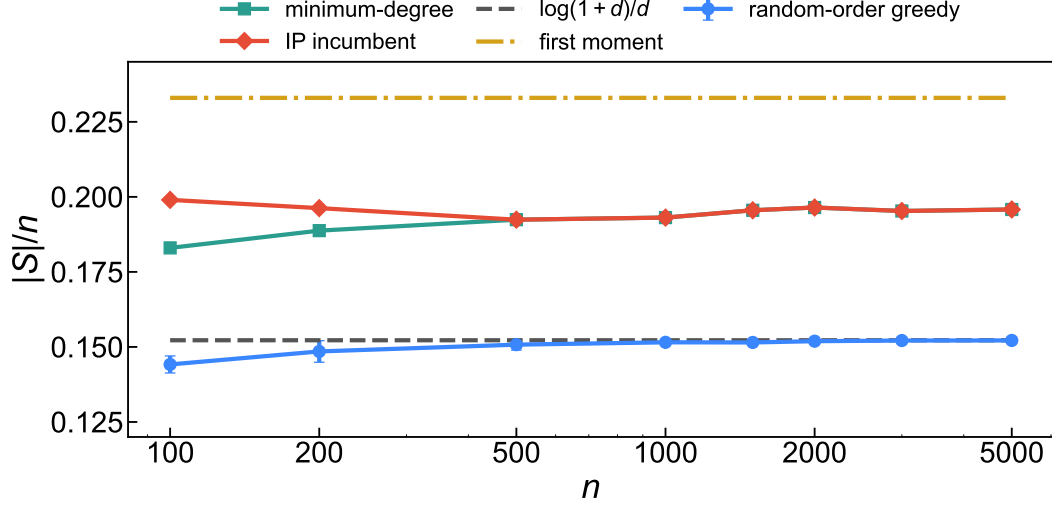


Figure 1: Feasible independent-set sizes found on $G(n, 20/n)$, normalized by n and averaged over seeds. Random-order greedy is the basic baseline, minimum-degree greedy is the warm start given to CBC, and the IP curve is CBC’s best feasible set at termination, with certification shown separately in Figure 2 and the CBC incumbent coinciding with the minimum-degree warm start from $n \geq 500$.

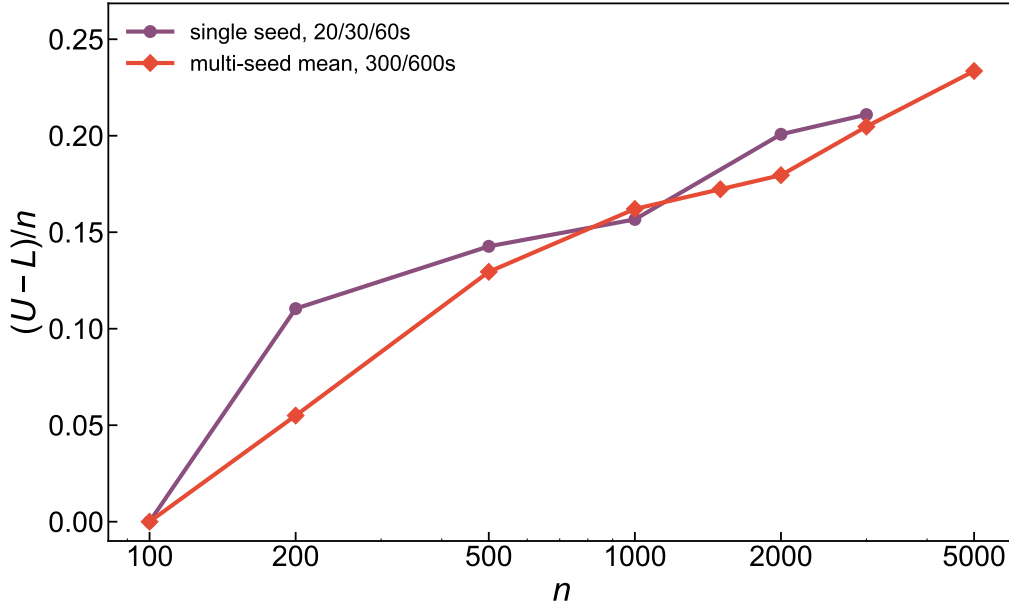


Figure 2: Certificate gap comparison between a single-seed random-graph experiment with 20/30/60-second limits and a multi-seed random-graph experiment with 300/600-second limits. The larger time budget reduces the gap for several sizes, while the larger random graphs still end with nonzero certificate gap.

The random-order greedy results agree closely with the finite- d greedy prediction, with mean density rising from 0.1442 at $n = 100$ to 0.1522 at $n = 5000$, essentially the value of $\log(1 + 20)/20$. Minimum-degree greedy is consistently larger, around 0.19–0.20 over the tested sizes, while CBC sometimes improves the warm start by a few vertices at $n = 200$ but still does not certify optimality.

From $n = 500$ onward, the recorded incumbent in the multi-seed experiment equals the warm start in every run.

Within the given time limit, the larger random graphs remain uncertified, as illustrated by the multi-seed experiment at $n = 3000$, where the mean incumbent density is 0.1953 and the mean converted upper-bound density is 0.4000. The shorter-and-longer time-limit comparison in Figure 2 shows that giving CBC more time closes part of the certificate gap, especially at the smaller sizes, while the gap remains nonzero on the larger random graphs. The random-graph experiment therefore separates the task of quickly obtaining a large feasible independent set through minimum-degree greedy from the harder task of closing the optimality gap with CBC at these sizes.

6 Real Network Experiment on SNAP ca-GrQc

We further implement the real-network instance on the SNAP ca-GrQc collaboration graph [5, 6]. It records coauthorship in the arXiv General Relativity and Quantum Cosmology category; a paper with k authors contributes a k -clique to the graph. SNAP reports 5242 nodes and 14496 raw edges. After symmetrizing duplicate rows, removing self-loops, and retaining vertices that become isolated after loop removal, our simple graph has

$$n = 5242, \quad m = 14484, \quad \bar{d} = \frac{2m}{n} = 5.526.$$

The graph has 355 connected components, with the largest component containing 4158 vertices. We solve the IP separately on each component and sum the component optima.

Table 2: SNAP ca-GrQc independent set results.

quantity	value
random greedy mean, 20 permutations	2179.20
random greedy standard deviation	15.17
random greedy best, 20 permutations	2202
minimum-degree greedy	2458
IP optimum	2459
IP certificate	exact
total IP time	35.1 seconds

The component-wise IP solve certifies

$$\alpha(G_{\text{ca-GrQc}}) = 2459.$$

Minimum-degree greedy obtains 2458, which is

$$\frac{2458}{2459} = 99.96\%$$

of the certified optimum. Random-order greedy is substantially weaker, with average performance

$$\frac{2179.20}{2459} = 88.62\%$$

of the optimum over 20 random permutations.

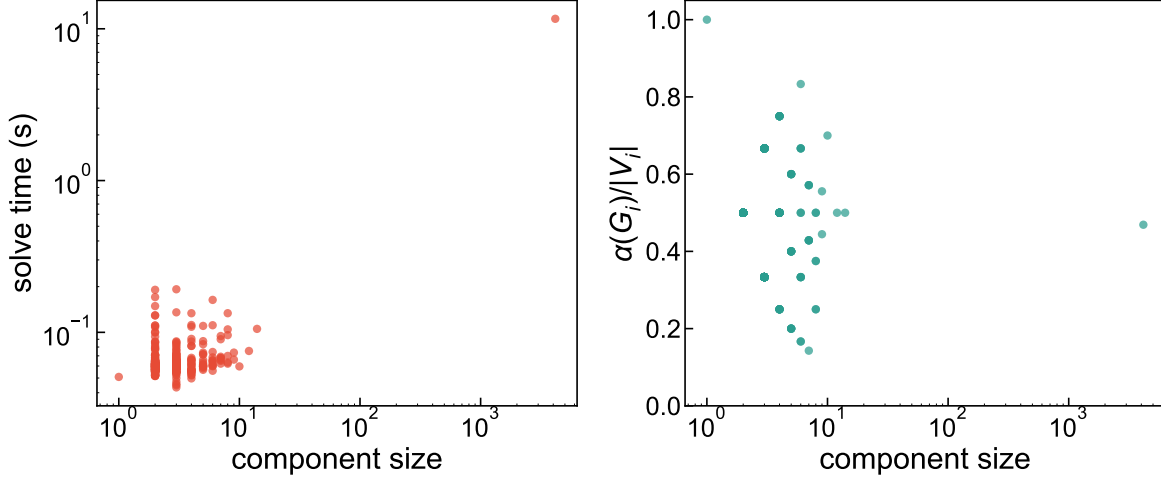


Figure 3: Component-level profile for ca-GrQc. The largest component is the dominant subproblem in solve time, while the smaller components contribute little computational cost and often have high independent-set density.

Substituting the observed average degree $\bar{d}$ into the Erdős–Rényi benchmarks gives

$$\begin{aligned} \frac{\log \bar{d}}{\bar{d}} n &= 1621.6, & \frac{\log(1 + \bar{d})}{\bar{d}} n &= 1779.4, \\ a_{\text{FM}} n &= 2625.5, & \frac{2 \log \bar{d}}{\bar{d}} n &= 3243.2. \end{aligned}$$

The certified optimum lies above the greedy-scale predictions and below the finite- d first-moment upper benchmark, although this comparison is only a null-model reference because the structural features of ca-GrQc differ significantly from those of an Erdős–Rényi graph with the same average degree.

Table 3: ca-GrQc and one matched Erdős–Rényi null instance generated with the same n and expected average degree $\bar{d}$.

graph	n	m	comps.	rand. mean	min-deg	L	U	time
ca-GrQc	5242	14484	355	2179.20	2458	2459	2459	35.1s
matched ER	5242	14782	—	1763.15	2115	2115	2605	32.1s

The matched Erdős–Rényi control uses $p = \bar{d}/n$. On this control instance, after the same 30-second budget, CBC has incumbent 2115 and converted upper bound 2605. The ca-GrQc instance, in contrast, is solved exactly after component decomposition. Therefore, this comparison establishes a narrow structural point that the advantage comes from the particular component structure and collaboration geometry of ca-GrQc, rather than from vertex count or density alone.

7 Discussion

The sparse random graph experiments highlight the distinction between incumbent quality and proof of optimality, since the primal side of the computation is strong enough for minimum-degree greedy to quickly find feasible independent sets above the random-order greedy scale, while the

dual side is much weaker and leaves CBC with large upper-bound gaps on the larger $G(n, 20/n)$ instances even though, in branch-and-bound terms, the solver is initialized with a good incumbent.

As Figure 2 shows, CBC improves the warm start on the smallest random instances and sometimes improves it at $n = 200$, although no $n = 200$ run is certified, and for $n \geq 500$ the multi-seed records show zero improvement over the warm start, so the empirical gain over random-order greedy at those sizes comes from the minimum-degree initialization.

The ca-GrQc experiment demonstrates a different use of the same IP formulation, where the graph’s 355 connected components reduce one 5242-vertex problem to many smaller subproblems and, together with the structure of the coauthorship network, make exact certification feasible. In this setting, the IP formulation is useful not only for storing a strong incumbent but also for proving optimality.

The comparison with random-graph formulas requires some care because, at $d = 20$, the first-moment benchmark is around $0.233n$, whereas the leading large- d expression $2 \log(d)n/d$ is around $0.300n$. For the random instances, the observed incumbents fall between the random-order greedy scale and this finite- d first-moment benchmark, while for the real network the Erdős–Rényi quantities are best viewed as baselines rather than predictions because ca-GrQc is far from an Erdős–Rényi graph.

8 Conclusion

On $G(n, 20/n)$, the warm-started IP pipeline returns feasible independent sets larger than the random-order greedy scale across the completed multi-seed experiment, although this success comes mostly from the minimum-degree greedy warm start. CBC certifies all $n = 100$ instances, but for $n \geq 200$ it does not certify optimality within the time limit, and for $n \geq 500$ it does not improve the warm start, so on the larger sparse random graphs the IP approach does not solve the problem in the exact optimization sense.

On the SNAP ca-GrQc collaboration network, the same formulation performs much better after connected-component decomposition, with CBC certifying

$$\alpha(G) = 2459,$$

while minimum-degree greedy obtains 2458 and random-order greedy averages 2179.20. The real-network experiment shows that IP can provide exact certificates on structured sparse networks even when it struggles to close optimality gaps on sparse Erdős–Rényi instances of comparable size.

Overall, random-order greedy matches the expected $\log(1 + d)n/d$ scale on the random graphs, minimum-degree greedy is substantially stronger, and IP is most valuable when it can convert a strong incumbent into a proof of optimality. In our experiments, that certification succeeds on ca-GrQc but not on the larger tested Erdős–Rényi graphs.

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